

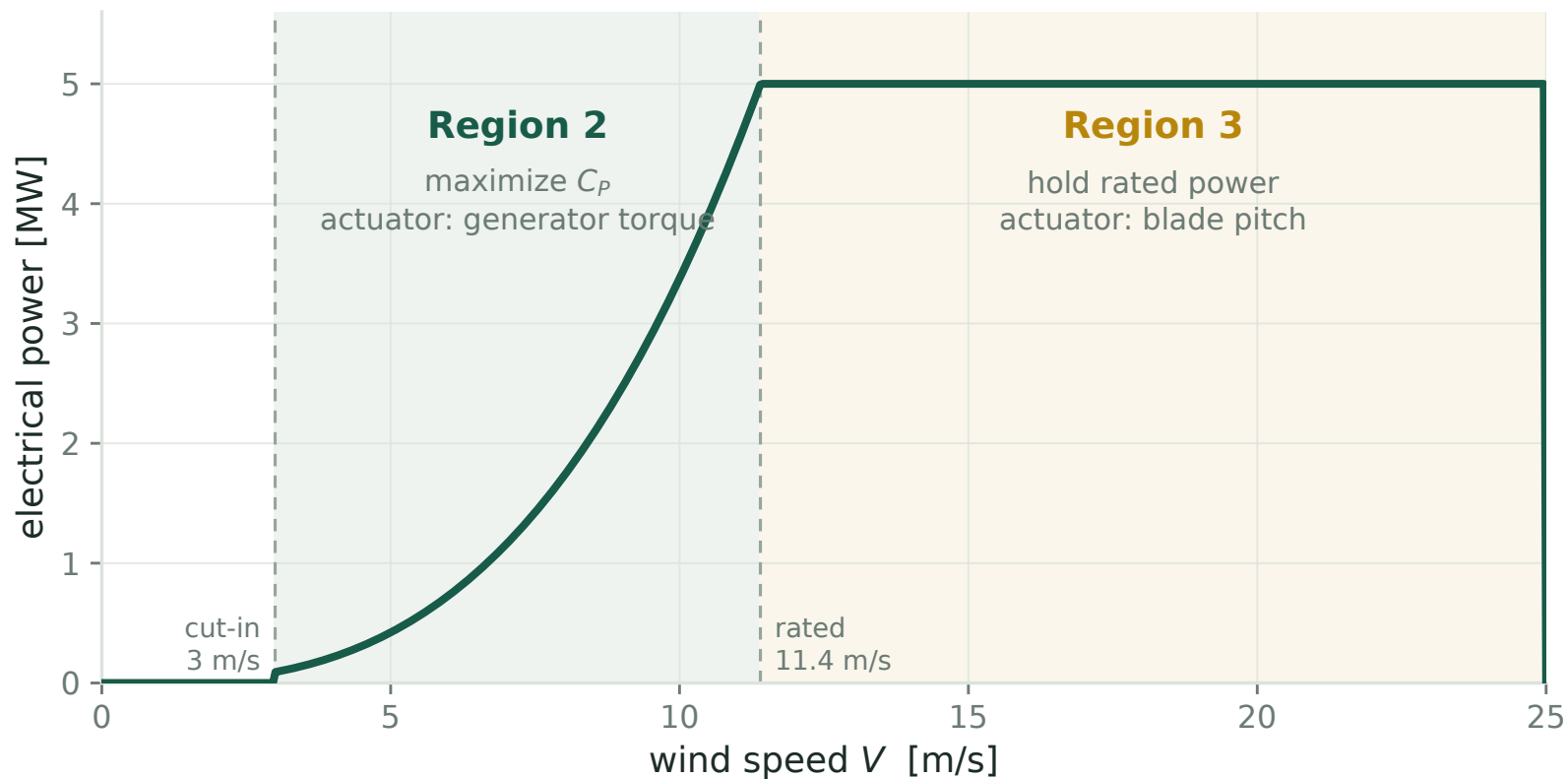
Extremum Seeking Control of Wind Turbines

The Region-2 problem, the extremum-seeking machinery built for it, and five open questions

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Two regimes, one optimization problem



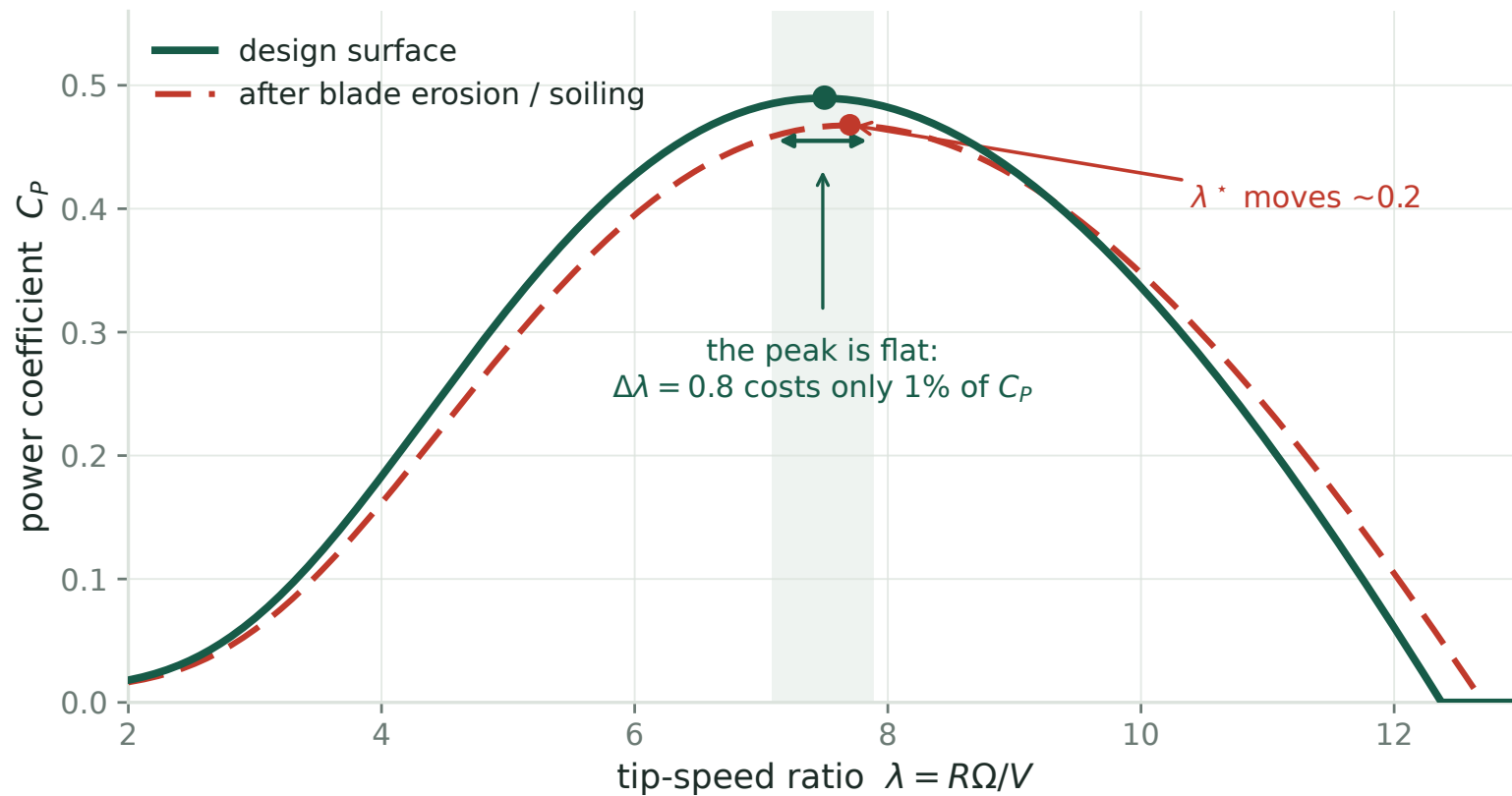
Region 3: above rated

There is more wind than the machine can use. Throw the excess away and hold rated power. A setpoint, an error signal and a PI controller.

Region 2: below rated

Take everything the wind offers. There is no setpoint, only an efficiency to maximize. **This is the only regime containing an extremum to seek.**

The hill



With the blades held at their design pitch, the rotor's efficiency depends on one number: the **tip-speed ratio** $\lambda = R\Omega/V$, blade tip speed divided by wind speed. Hold λ at λ^* and you capture the most power available.

You cannot measure either axis

You measure

Ω rotor speed: encoder, exact
 τ_g generator torque: you command it
 P_g generator power: a wattmeter



The curve is drawn in

$\lambda = R\Omega/V$ horizontal axis
 $C_p = P/(\frac{1}{2}\rho AV^3)$ vertical axis
 $\partial C_p / \partial \lambda$ the slope, from both

all three need V

So you could try to build the curve directly, from $\hat{V}$. Both axes carry the same unknown, so an error there does not scatter the points: it rescales the curve and carries the peak along, $\hat{\lambda}^* = \lambda^* / c$. Resolving the 0.2 that erosion produces would need $\hat{V}$ good to 3%. Extremum seeking does not go this way at all.

Both coordinates of the hill are written in terms of V , the wind speed actually reaching the rotor, which is exactly what a turbine cannot measure. **So this is the route extremum seeking declines to take.**

Sitting on a hill you cannot see

How the controller finds the peak without measuring where it is.

The $k\Omega^2$ law

The rotor is one rigid body, so what governs it is

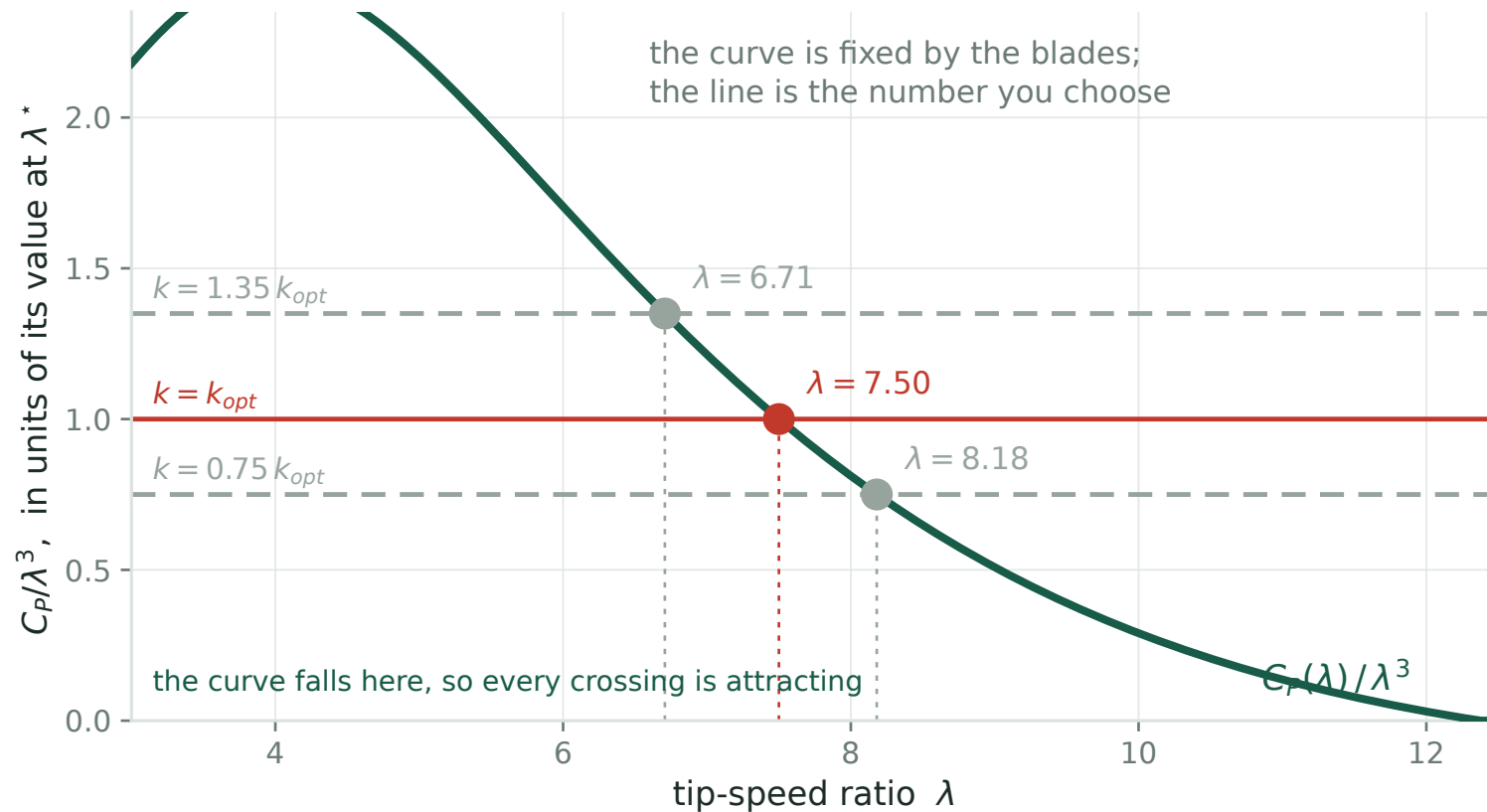
$$I\dot{\Omega} = \tau_{\text{aero}} - \tau_g, \quad \tau_{\text{aero}} = \frac{1}{2}\rho\pi R^3 V^2 C_Q(\lambda), \quad \tau_g = k\Omega^2$$

Put $\dot{\Omega} = 0$, substitute $\Omega = \lambda V/R$, and divide through by V^2 . Both V and Ω disappear, leaving a condition on λ alone:

$$\boxed{\frac{C_P(\lambda)}{\lambda^3} = \frac{2k}{\rho\pi R^5}}$$

So $\dot{\Omega} = 0$ does not leave the speed free to have drifted anywhere. The left side is a **function** of λ whose shape is set by the blade geometry; the right side is a genuine constant, the one you pick with k . The rotor can only rest where the two meet.

Picking k picks a resting point



Raise k and the line rises, so the crossing slides left to a lower λ . Nothing here needed the wind speed, and nothing needed a measurement.

Why the rotor goes to that root

Resting there is not the same as being drawn there. Write the net torque as a function of the state, $I\dot{\Omega} = F(\Omega)$ with

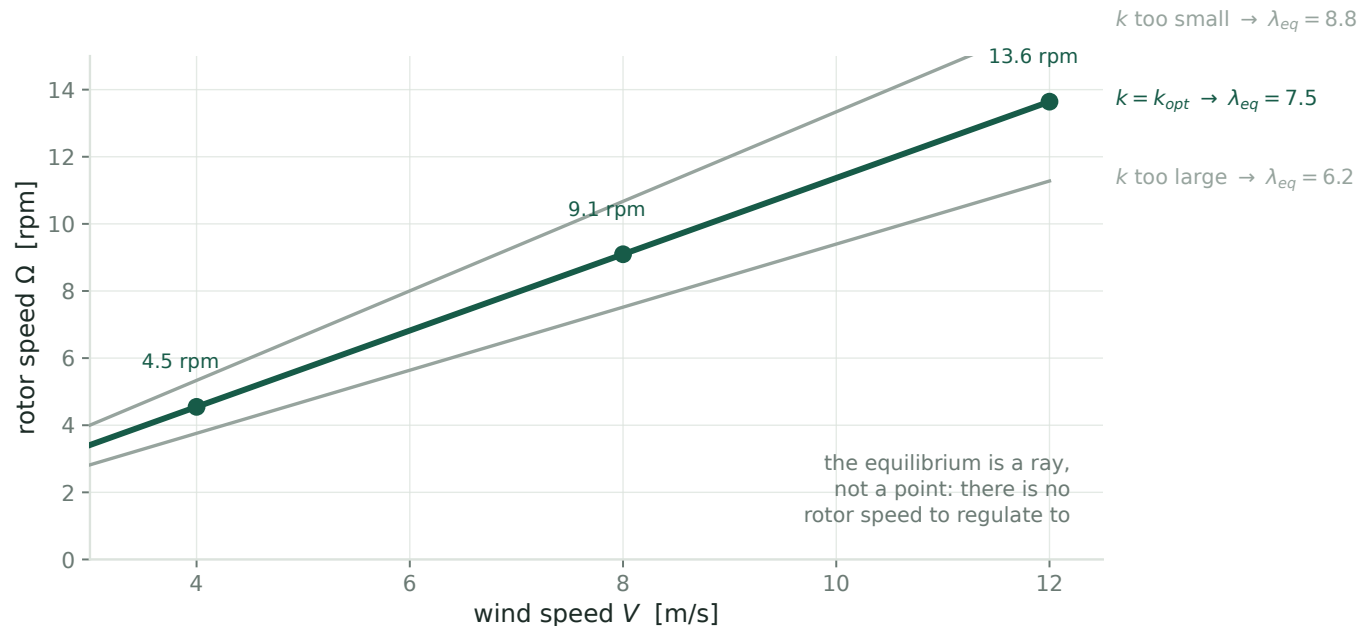
$$F(\Omega) = \frac{1}{2}\rho\pi R^3 V^2 C_Q\left(\frac{R\Omega}{V}\right) - k\Omega^2$$

A small perturbation $\delta = \Omega - \Omega_{eq}$ obeys $I\dot{\delta} = F'(\Omega_{eq})\delta$, so it decays exactly when $F'(\Omega_{eq}) < 0$. Substituting the equilibrium relation collapses that to a statement about the curve alone:

$$F'(\Omega_{eq}) < 0 \quad \Longleftrightarrow \quad \frac{d}{d\lambda}\left(\frac{C_P}{\lambda^3}\right) < 0 \quad \Longleftrightarrow \quad \frac{d \ln C_P}{d \ln \lambda} < 3$$

At the peak $C'_P = 0$, so the middle expression is $-3C_P^{\max}/\lambda^4$, negative for **any** turbine. The design point is always attracting, whatever the blades look like.

There is no rotor speed to regulate to



The crossing gives $\lambda_{eq}(k)$, with no V in it. Convert back through $\lambda = R\Omega/V$:

$$\Omega_{eq}(V) = \frac{\lambda_{eq}(k)}{R} V$$

So **the crossing is this ray**, in rotor-speed coordinates: one point there, one line here, every point on it the same λ . The controller never rejects the wind. It makes that line an attracting manifold and lets the wind slide the machine along it.

So the problem reduces to one number

What we have

A single scalar knob k , and a monotone map from k to the tip-speed ratio the machine settles at.

What we want

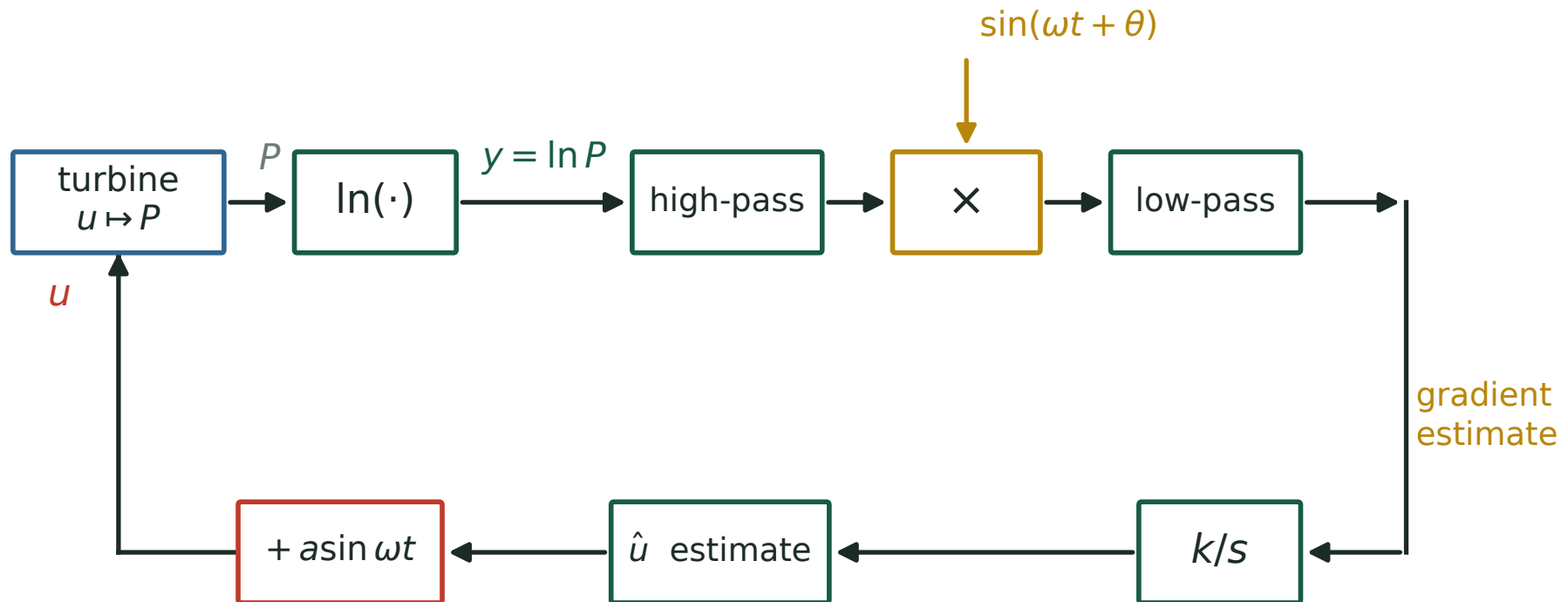
The k that lands us on the peak. The factory value is wrong at commissioning and gets worse as the blades age.

Find the maximum of a function you cannot evaluate, by turning a knob and watching a noisy meter. That is **extremum seeking control**, and it is what the last decade of work at UTD has been about.

Extremum seeking

Estimate a derivative you cannot measure by wiggling the input.

The loop



The gradient is never measured. It is inferred from how much of the injected tone comes back out.

Add a slow sinusoid $a \sin \omega t$ to the knob. If the output wiggles **in phase**, you are on the uphill side; **out of phase**, the downhill side. Multiply and average, and what survives is proportional to the slope.

The one identity that made it practical

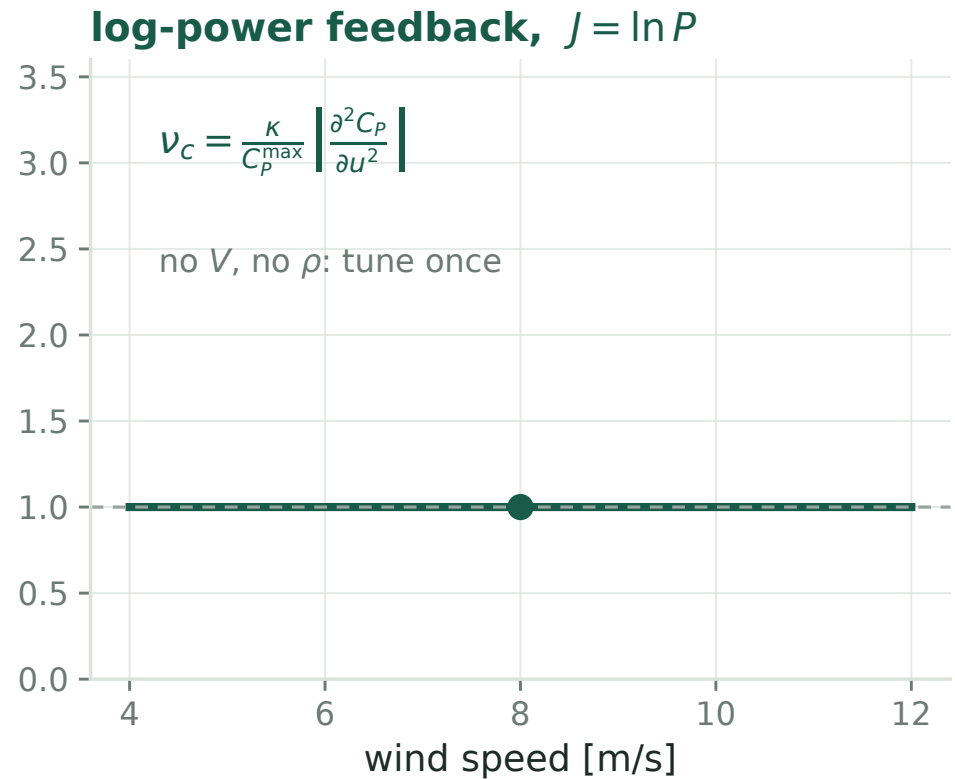
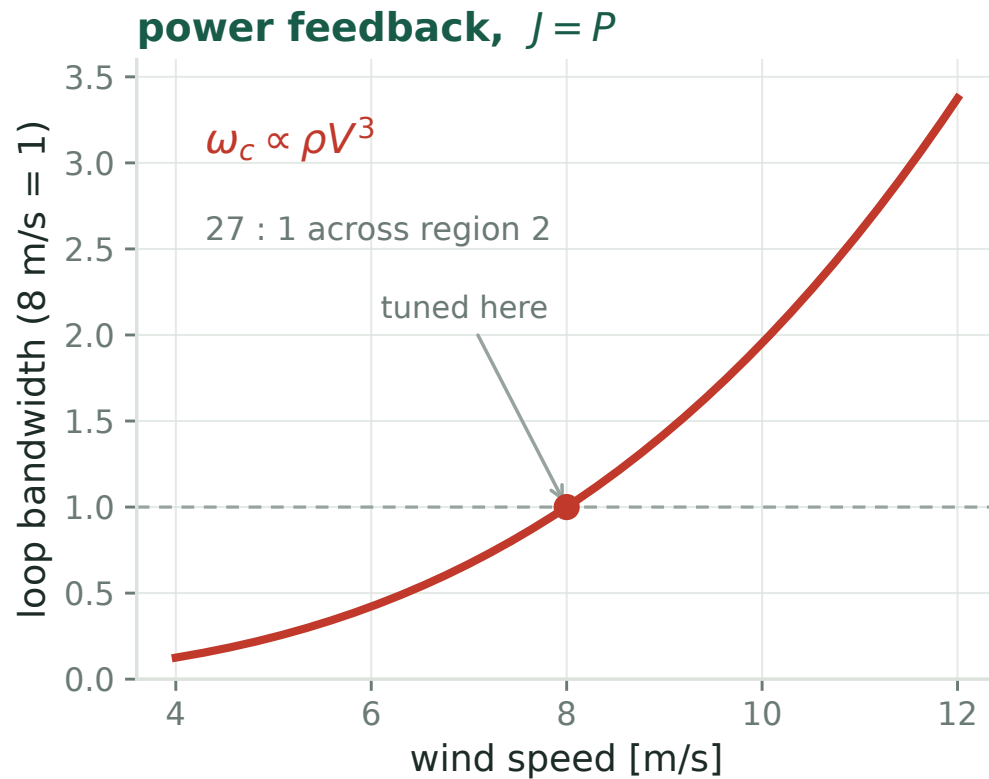
Power is proportional to the cube of the wind speed, so the gradient of power is too, and the wind varies by 3:1 across Region 2. A loop tuned at 8 m/s is 27 times too slow at 4 m/s and 27 times too aggressive at 12 m/s.

Rotea's 2017 observation is one line. Since $\ln P = \ln(\frac{1}{2}\rho A) + 3 \ln V + \ln C_P(u)$, and the wind term does not depend on u :

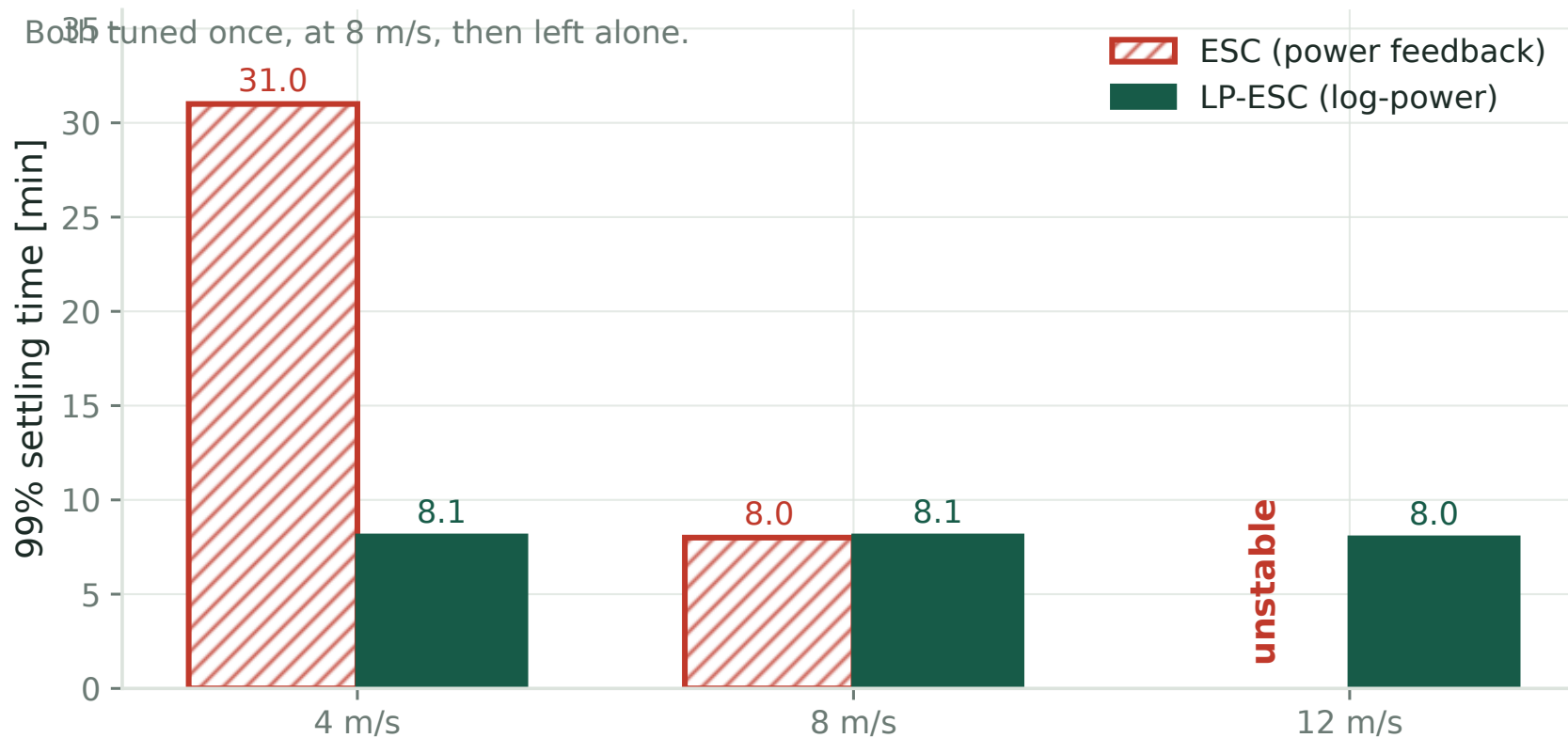
$$\frac{\partial \ln P}{\partial u} = \frac{1}{C_P} \frac{\partial C_P}{\partial u}$$

Feed the algorithm $\ln P$ instead of P and the loop bandwidth becomes $\nu_c = (\kappa/C_P^{\max}) |\partial^2 C_P / \partial u^2|$, free of both the wind speed and the air density. **Tune it once; it works everywhere.**

What the logarithm fixes



It holds up in high-fidelity simulation



Ciri, Leonardi & Rotea 2019

Large-eddy simulation of the full flow around an NREL 5 MW rotor, with shear and 10% turbulence. No analytic gradient anywhere.

The result

Log-power holds 8 minutes across a 3:1 wind range. Plain ESC is four times slower at the bottom and **goes unstable** at the top.

Making it fast: LP-PIESC

LP-ESC

inject $a \sin \omega t$, multiply the response by $\sin(\omega t + \theta)$, low-pass
 $\Rightarrow$ one gradient per dither period



LP-PIESC

fit $\dot{y} = \theta_0 + \theta_1(u - \hat{u})$ by recursive least squares; θ_1 is the gradient. Add a proportional term.

Dither amplitude

a cut to 25%
of the LP-ESC value

Energy vs. a controller
that already knows k_{opt}

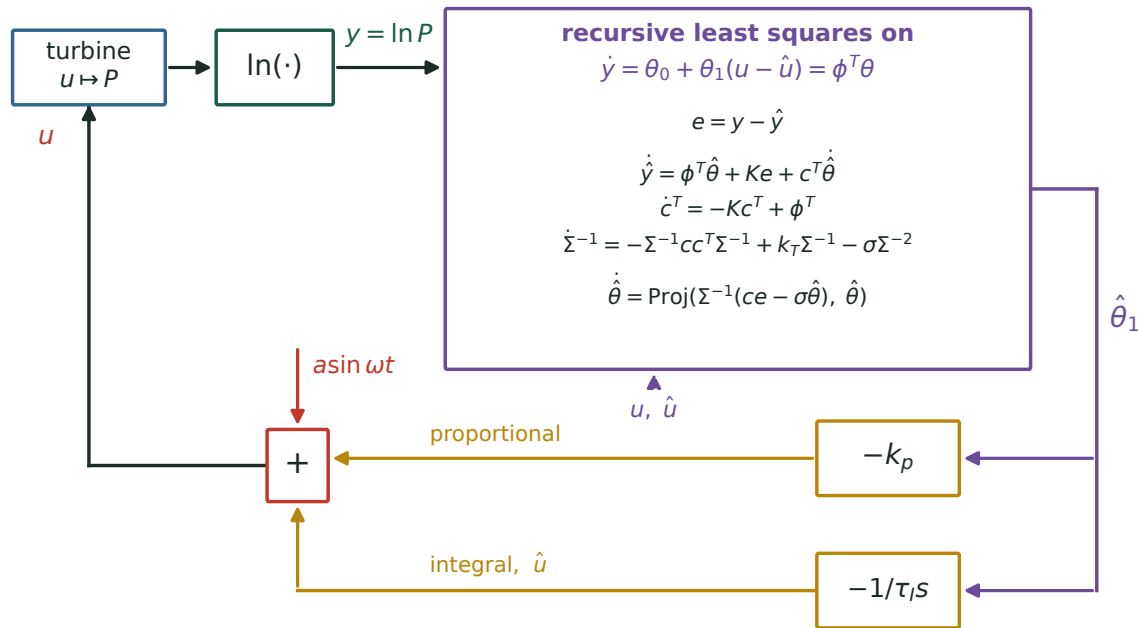
LP-ESC -14.2%
LP-PIESC -0.3%

Wake steering, wind tunnel
12 turbines, yaw

+8.9% farm power,
converged in seconds

Replace demodulation with **recursive least squares** on $\dot{y} = \theta_0 + \theta_1(u - \hat{u})$ and add a proportional term. The gradient θ_1 now comes with a covariance matrix Σ that the algorithm already propagates, and that none of these papers reports.

LP-PIESC in full



Compare slide 12. The high-pass, the multiplier, the low-pass and the $\sin(\omega t + \theta)$ reference are gone, replaced by the one block above. The k/s integrator has become a PI pair. Everything else is unchanged.

The whole demodulation chain is gone: no multiplier, no high-pass, no low-pass, no phase angle. The gradient is $\hat{\theta}_1$, read directly out of a least-squares fit to $\dot{y} = \theta_0 + \theta_1(u - \hat{u})$.

What the estimator is fitting

The local model is affine in the distance from the current estimate:

$$\dot{y} = \theta_0 + \theta_1 (u - \hat{u}) = \phi^T \theta, \quad \phi = \begin{bmatrix} 1 \\ u - \hat{u} \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}$$

so ϕ and θ are **two-vectors**, and θ_1 is the gradient we are after. Everything else in the box has a shape forced by these:

vectors in $\mathbb{R}^2$

$\phi, \theta, \hat{\theta}$, and c

scalars

$y, \hat{y}, e$, and the constants K, k_T, σ

2×2 matrices

Σ and Σ^{-1}

The obstacle: the model is written in $\dot{y}$, and $\dot{y}$ is not measured. Differentiating a turbulent power signal to get it would be hopeless.

Why there is a filtered regressor

Predict y with $\dot{y} = \phi^T \hat{\theta} + Ke + c^T \dot{\hat{\theta}}$ and let $\tilde{\theta} = \theta - \hat{\theta}$. Subtracting from $\dot{y} = \phi^T \theta$,

$$\dot{e} = \phi^T \tilde{\theta} - Ke - c^T \dot{\hat{\theta}}$$

Now define $\eta = e - c^T \tilde{\theta}$ and differentiate, using $\dot{\tilde{\theta}} = -\dot{\hat{\theta}}$ and $\dot{c}^T = -Kc^T + \phi^T$:

$$\dot{\eta} = \underbrace{\phi^T \tilde{\theta} - Ke - c^T \dot{\hat{\theta}}}_{\dot{e}} - \underbrace{(-Kc^T + \phi^T) \tilde{\theta}}_{\dot{c}^T \tilde{\theta}} + \underbrace{c^T \dot{\hat{\theta}}}_{-c^T \dot{\tilde{\theta}}} = -K(e - c^T \tilde{\theta}) = -K\eta$$

Everything cancels but $-K\eta$. So $\eta \rightarrow 0$ exponentially, which means $e \rightarrow c^T \tilde{\theta}$: the **measurable** error becomes the regressor c dotted with the parameter error. The odd $c^T \dot{\hat{\theta}}$ term in the predictor is there for exactly this, since it is what cancels $c^T \dot{\tilde{\theta}}$.

Which makes it a ridge regression

Rearranging $e = c^T \theta - c^T \hat{\theta}$ and defining $z := e + c^T \hat{\theta}$, we get $z = c^T \theta$: known regressor, computable observation, unknown parameter. Fit it with forgetting **and** a penalty on the size of $\hat{\theta}$:

$$J(t, \hat{\theta}) = \int_0^t e^{-k_T(t-s)} \left[|z - c^T \hat{\theta}|^2 + \sigma |\hat{\theta}|^2 \right] ds$$

Setting $\partial J / \partial \hat{\theta} = 0$ gives the normal equations $\Sigma \hat{\theta} = b$, where, with $w = e^{-k_T(t-s)}$,

$$\Sigma = \underbrace{\int_0^t w c c^T ds}_A + \sigma \underbrace{\int_0^t w ds I}_m, \quad b = \int_0^t w c z ds$$

Σ is the information matrix of a **ridge**-regularized least squares, and it is the Σ of the block diagram. The next slide shows that both places σ appears in the algorithm come from the one penalty written here.

One cost, two equations

The Σ recursion. Since $\dot{A} = cc^T - k_TA$ and $\dot{m} = 1 - k_Tm$,

$$\dot{\Sigma} = (cc^T - k_TA) + \sigma(1 - k_Tm)I = cc^T - k_T \underbrace{(A + \sigma m I)}_{\Sigma} + \sigma I$$

The parameter update. Differentiate $\hat{\theta} = \Sigma^{-1}b$, substitute $\dot{\Sigma}$ and $\dot{b} = cz - k_Tb$, and the k_T terms cancel:

$$\dot{\hat{\theta}} = -\Sigma^{-1}\dot{\Sigma}\hat{\theta} + \Sigma^{-1}\dot{b} = \Sigma^{-1}c(z - c^T\hat{\theta}) - \sigma\Sigma^{-1}\hat{\theta} = \Sigma^{-1}(ce - \sigma\hat{\theta})$$

That is why c multiplies e , and why the same σ appears twice. It is not a robustness patch bolted on in two places: it is one ridge penalty, showing up once through Σ and once through the shrinkage of $\hat{\theta}$.

So why is demodulation still the default?

What LP-PIESC assumes

That $\dot{y}$ is affine in $(u - \hat{u})$, that the regressor stays persistently exciting, and that you can bound the parameters in advance for the projection.

What demodulation assumes

That the plant is roughly static at one frequency and the peak is locally unimodal. No model structure at all.

And the count goes the wrong way. LP-ESC has six parameters: a, ω, θ , two cutoffs, k . LP-PIESC has seven: $a, \omega, k_T, K, \sigma, k_p, \tau_I$, and Rotea's 2024 multi-row paper says all but two were set by trial and error.

Which is exactly the complaint we were given. The fast algorithm is the one that “takes too long to tune its parameters.”

Where things stand

Solved

Consistency across wind speed. The logarithm removes the V^3 dependence exactly, in theory, in LES, and in the tunnel.

Solved

Speed. LP-PIESC converges fast enough that the search costs 0.3% of energy rather than 14%.

Not solved

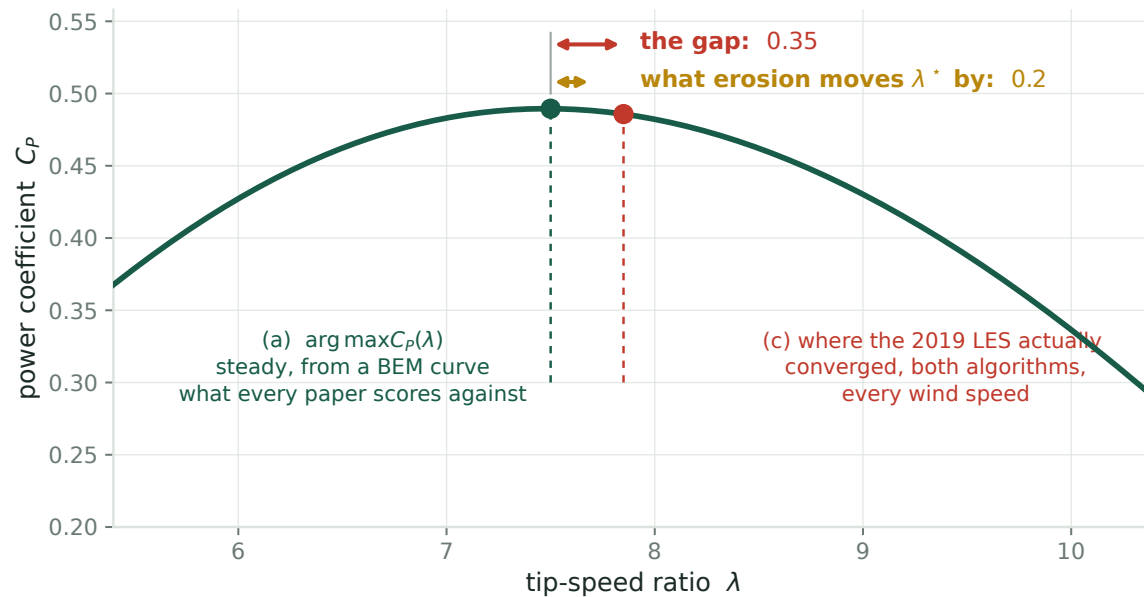
Everything to do with **noise**. “The enemy is process noise”, and none of these papers reports a variance.

Extremum seeking works well when the process noise is negligible. Otherwise you must average, and averaging costs convergence time. That trade-off has not been quantified for this problem.

Five questions, audited

Two of them closed under inspection. Q1 turned out not to be a control problem and Q2 was fixed in 2017. What is left is Q3, Q4 and Q5.

Q1 · Where does the torque gain converge?



offset from $\lambda^* = 7.5$

[CLR19] Table 4, shear +10% TI +0.22 to +0.35

[CLR19] Table 3, uniform inflow +0.10 to +0.25

finite-difference bias, computed -0.013

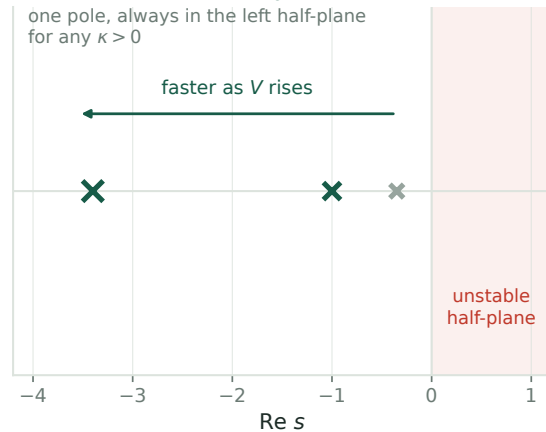
turbulent peak shift, computed +0.005 to +0.037

Most of the gap is already there in inflow with **no turbulence**, where neither computed candidate can act, and the scatter is bounded at $\sigma_\lambda \lesssim 0.14$. What is left is the reference: design λ^* is a blade-element number, the LES rotor is actuator-line. **Closed as a control question, open as a modeling one.** Worked through in the Q1 deck.

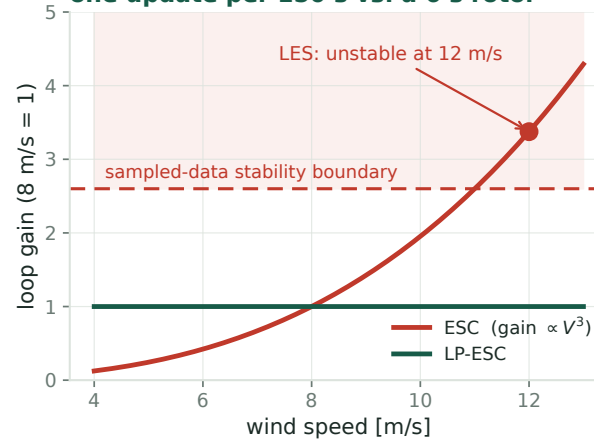
Q2 · What sets the stability limit?

2017 model: $\dot{\tilde{u}} = -\omega_c \tilde{u}$

one pole, always in the left half-plane for any $\kappa > 0$



one update per 150 s vs. a 6 s rotor



[R17] Eq. 8 reduces the loop to $\dot{\tilde{u}} = -\omega_c \tilde{u}$ with $\omega_c \propto \rho V^3$. One pole, negative real, for **any** gain: it cannot go unstable. [CLR19] report plain ESC unstable at 12 m/s, in uniform inflow and again in turbulence.

[R17] p. 4506 names the effect and cannot model it: “an increase in crossover frequency implies a decrease of the stability margins.” A first-order lag has no margin to lose. The missing element is that the loop is *sampled*: one update per 150 s dither period, so $\tilde{u}_{n+1} = (1 - K)\tilde{u}_n$ with $K = \kappa_{\text{eff}}|J''|$, giving $0 < K < 2$ where the continuous model gives no bound at all. Since $K \propto V^3$, high wind walks across it.

Closed, and closed by the existing fix. The logarithm removes the V^3 exactly, so LP-ESC holds one K at every wind speed. Nor is $K < 2$ a speed ceiling: $K = 1$ is deadbeat in one update. What survives is that $K = \kappa_{\text{eff}}|J''|$ still needs a curvature nobody measures, which is Q4.

Q3 · Does the turbine need to probe at all?

Two things are true and encouraging. τ_{aero} is **observable from the encoder alone**, no wind sensor and no C_P : augment it as a state on $I_{\text{eff}}\dot{\Omega} = \tau_{\text{aero}} - \tau_g$ and $[C; CA]$ has rank 2. And turbulence already moves λ about as far as the probe does, $\sigma_\lambda = 0.41$ against 0.34, verified in simulation.

Neither helps. Given a candidate curve, each instant supplies **one equation and one unknown** $V(t)$, so the curve is unconstrained pointwise. The degeneracy is not a direction, it is the whole function space.

posited curve	solvable	implied mean V	implied TI	corr. with true V
true	100%	8.00	9.3%	1.0000
peak flattened 30%	100%	7.99	9.3%	0.9999

A curve that is simply **wrong** reproduces the record exactly, with an implied wind indistinguishable from the real one. **This is why extremum seeking exists.** A known excitation is not a convenience; it is the only thing that breaks the degeneracy. The answer to Q3 is yes, and structurally so.

Q4 · What the curvature is doing

Look again at the result of Rotea (2017):

$$\nu_c = \frac{\kappa}{C_P^{\max}} \left| \frac{\partial^2 C_P}{\partial u^2}(u_o) \right|$$

Choosing the gain κ to hit a target settling time **requires the curvature at the peak**, currently taken from a design curve that the LES says is wrong by 10% in C_P . The gain calibration is not self-consistent, and “it takes too long to tune” is partly a symptom of that.

The curvature also sets the terminal scatter, $\sigma_\lambda \approx \sigma_{\hat{g}}/|H|$, and the energy lost to mistuning is quadratic in σ_λ . The same number sets the gain, the achievable accuracy and the cost of missing.

Q4 · Two routes to the same covariance

In stochastic approximation the asymptotic covariance is $\Sigma^{PR} = A^{-1} \Sigma_W A^{-T}$,
and for a gradient method $A = \nabla^2 \Gamma(\theta^*)$.
The Hessian is in the answer whether or not you estimate it.

Zap SA (Meyn)

$$\begin{aligned}\hat{A}_{n+1} &= \hat{A}_n + \varepsilon_{n+1} [A_{n+1} - \hat{A}_n] \\ \theta_{n+1} &= \theta_n + \alpha_{n+1} G_{n+1} f_{n+1} \\ G_{n+1} &= -[\varepsilon I + \hat{A}^T \hat{A}]^{-1} \hat{A}\end{aligned}$$

Estimate the Jacobian on a fast timescale, invert it, use it as a gain.



Polyak-Ruppert averaging

$$\theta_N^{PR} = \frac{1}{N - N_0} (\theta_{N_0} + \dots + \theta_N)$$

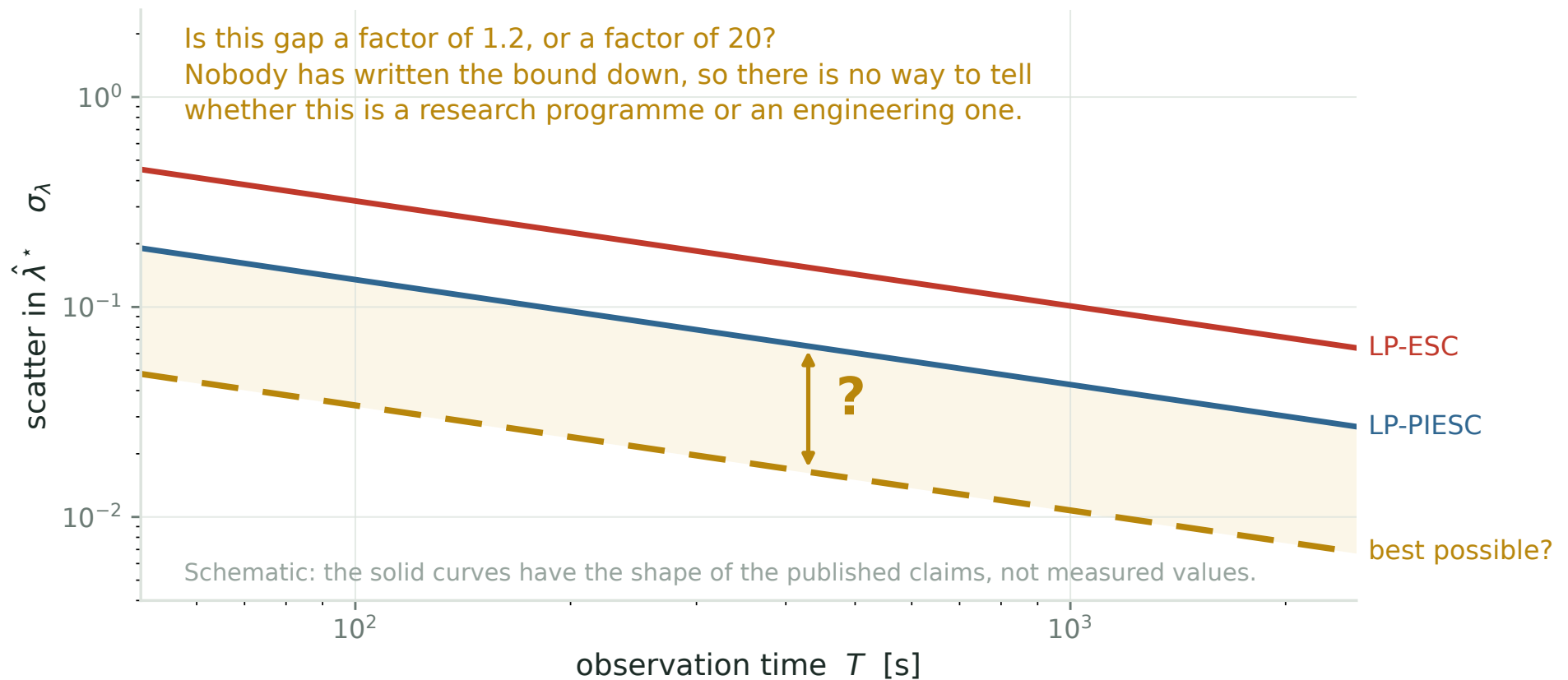
Average the iterates.
No curvature estimate anywhere.



Both attain the same optimal covariance. Meyn: “Given the simplicity of the recursions defining θ^{PR} , this is probably the first algorithm to try in most applications.”

Meyn does not estimate the Hessian in the ESC papers. It appears in the *analysis*, as $A = \nabla^2 \Gamma(\theta^*)$ in the asymptotic covariance. Where he does estimate it, in Zap SA, it is a two-timescale Jacobian estimate used as a matrix gain.

Q5 · How good could any algorithm be?



Given a turbulence spectrum, an observation time and a fatigue budget, how well can **anyone** locate λ^* ? Nobody has written the bound down.

Why turbulence is not ordinary noise

In classical stochastic approximation

Noise is pure nuisance. It corrupts the measurement and you average it away.

Here

Turbulence is **simultaneously the noise and the excitation**. It is what corrupts the gradient estimate *and* what sweeps the rotor across the curve for free.

Both sides are now measured, and they are the same size:

	σ_λ contributed
turbulence, through the rotor high-pass at 0.029 Hz	0.41
the probe, at the published $a = 13.6\%$ of k_{opt}	0.34

The excitation the wind supplies for free is **comparable to the one the controller pays for**. That is what made Q3 worth asking. It still fails, but on identifiability rather than on signal strength: an unknown wind absorbs any curve you propose, so excitation of unknown magnitude cannot substitute for excitation you chose.

The experiment I would run first

Open the loop. Hold the torque gain at a series of fixed values, run each long enough to average the turbulence, and measure the **simulator's own** $C_P(\lambda)$. Then ask where its peak is.

Why this one

Every published offset is scored against a blade-element $\lambda^* = 7.5$. Q1 showed the offset survives in inflow with no turbulence, so the reference is the only candidate left. This measures it directly.

What it costs

No new machinery, no closed loop, a few days of compute. And the same runs give σ_λ at each gain, which is the K_v nobody can currently predict.

Note the earlier obvious experiment, sweeping the dither amplitude to look for an a^2 scaling in the offset, is **not** worth running: the predicted effect is 0.013 against a seed spread larger than that.

Where this leaves things

The algorithms work. What is missing is the statistics of the quantity they compute.

What exists

A model-free loop, consistent across wind speed, with results from LES, a wind tunnel and a field test.

What does not

A variance for the gradient estimate, a reference the simulations can honestly be scored against, and a performance bound.

verdict	
Q1 estimand	closed; the offset is in the reference, not the algorithm. A measurement, not a program
Q2 stability limit	closed; sampling explains it and the ln of 2017 already fixes it
Q3 is the dither needed	closed; an unknown wind absorbs any curve, so a known probe is the only thing that identifies one
Q4 matrix gain	open, and simulated. The curvature does leave the loop gain
Q5 performance bound	untouched

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